# LIBRARY MANAGEMENT SYSTEM

Assignment 1 – Python Program with Flowchart and Output
Roll Number: 31

## 1. Python Code

**class Book:**

```
    def __init__(self, isbn, title, author):
        self.isbn = isbn
        self.title = title
        self.author = author
        self.is_issued = False
        self.issued_to = None

    def display(self):
        if self.is_issued:
            status = f"Issued to {self.issued_to}"
        else:
            status = "Available"

        print(f"ISBN : {self.isbn}")
        print(f"Title: {self.title}")
        print(f"Author: {self.author}")
        print(f"Status: {status}")
        print("-" * 30)


class Patron:
    def __init__(self, pid, name):
        self.pid = pid
        self.name = name

    def display(self):
        print(f"ID   : {self.pid}")
        print(f"Name : {self.name}")
        print("-" * 30)


class Library:
    def __init__(self):
        self.book_list = []
        self.patron_list = []
```

```
def find_book(self, isbn):
    for book in self.book_list:
        if book.isbn == isbn:
            return book
    return None

def find_patron(self, pid):
    for patron in self.patron_list:
        if patron.pid == pid:
            return patron
    return None

def add_book(self):
    isbn = input("Enter ISBN: ")

    if self.find_book(isbn):
        print("Book already exists.")
        return

    title = input("Enter Title: ")
    author = input("Enter Author: ")

    self.book_list.append(Book(isbn, title, author))
    print("Book added successfully.")

def register_patron(self):
    pid = input("Enter Patron ID: ")

    if self.find_patron(pid):
        print("Patron already registered.")
        return

    name = input("Enter Name: ")
    self.patron_list.append(Patron(pid, name))
    print("Patron registered successfully.")

def view_data(self):
    print("\n===== BOOKS =====")

    if len(self.book_list) == 0:
        print("No books available.")
    else:
        for book in self.book_list:
            book.display()

    print("\n===== PATRONS =====")

    if len(self.patron_list) == 0:
        print("No patrons registered.")
    else:
        for patron in self.patron_list:
            patron.display()

def borrow_book(self):
    isbn = input("Enter ISBN: ")
    pid = input("Enter Patron ID: ")

    book = self.find_book(isbn)
```

```
        patron = self.find_patron(pid)

        if book is None:
            print("Book not found.")
            return

        if patron is None:
            print("Patron not found.")
            return

        if book.is_issued:
            print("Book is not available.")
        else:
            book.is_issued = True
            book.issued_to = patron.name
            print("Book issued successfully.")

    def return_book(self):
        isbn = input("Enter ISBN: ")
        pid = input("Enter Patron ID: ")

        book = self.find_book(isbn)
        patron = self.find_patron(pid)

        if book is None or patron is None:
            print("Invalid details.")
            return

        if book.is_issued and book.issued_to == patron.name:
            book.is_issued = False
            book.issued_to = None
            print("Book returned successfully.")
        else:
            print("This patron has not borrowed the book.")


def menu():
    print("\n========== LIBRARY MANAGEMENT ==========")
    print("1. Add Book")
    print("2. Register Patron")
    print("3. View Books & Patrons")
    print("4. Borrow Book")
    print("5. Return Book")
    print("6. Exit")


def main():
    library = Library()

    while True:
        menu()

        choice = input("Enter your choice: ")

        if choice == "1":
            library.add_book()

        elif choice == "2":
```

```
            library.register_patron()

        elif choice == "3":
            library.view_data()

        elif choice == "4":
            library.borrow_book()

        elif choice == "5":
            library.return_book()

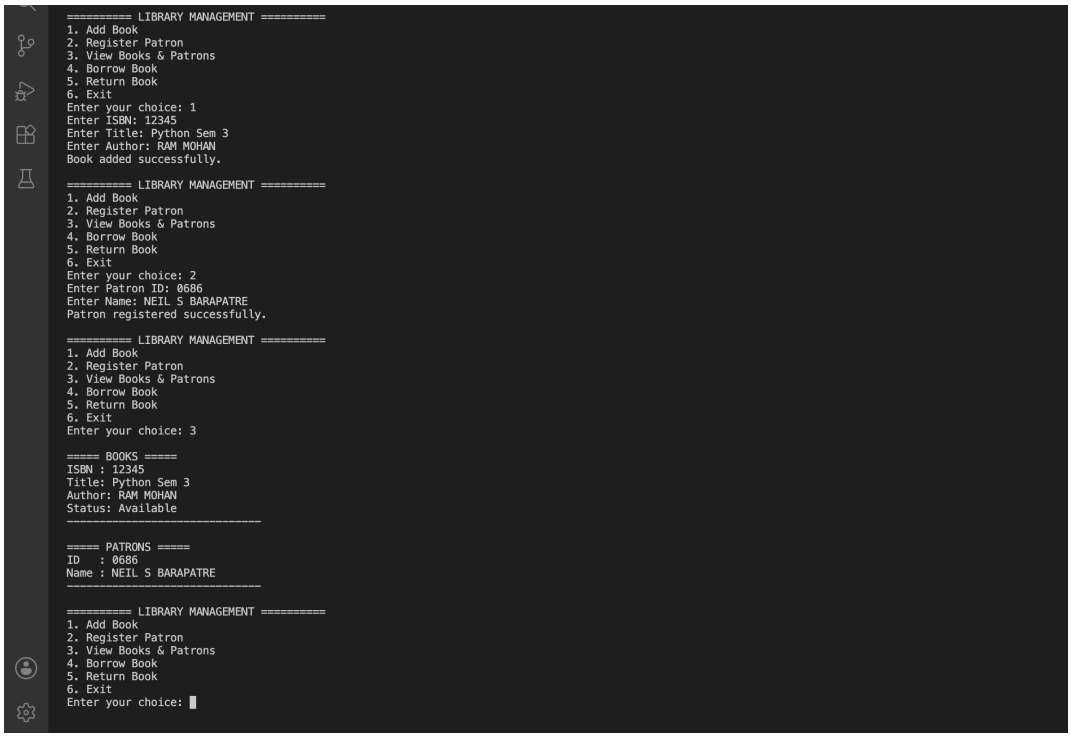
        elif choice == "6":
            print("Thank you for using Library Management System.")
            break

        else:
            print("Invalid choice. Try again.")

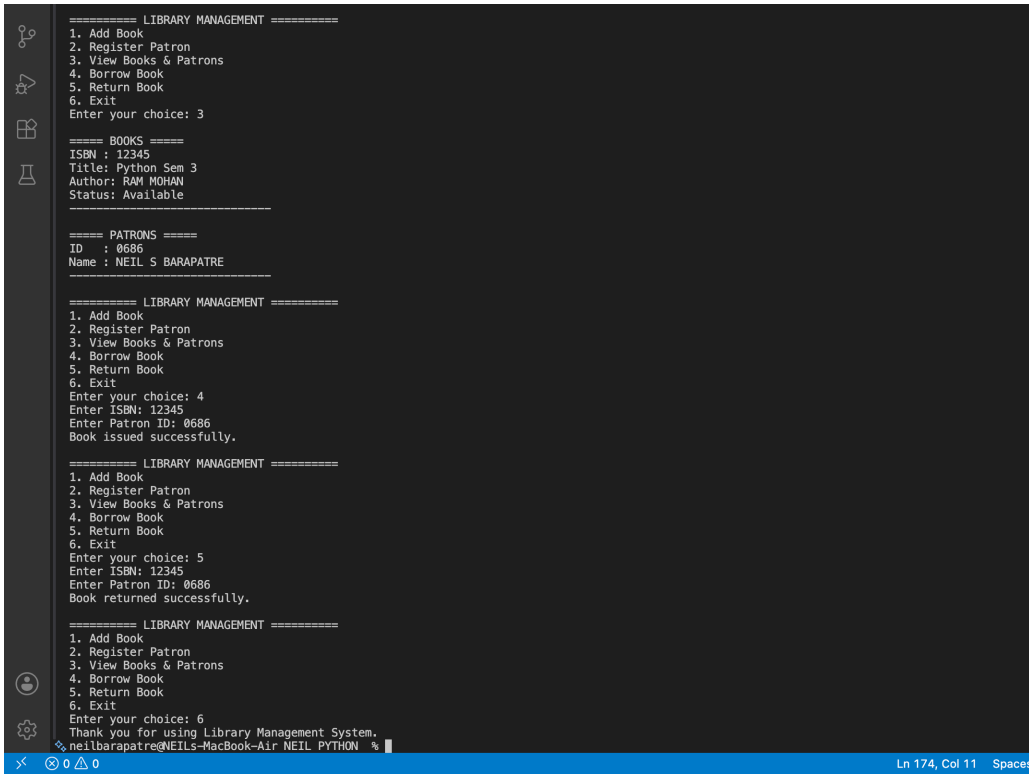

if __name__ == "__main__":
    main()
```

# 2. Program Output

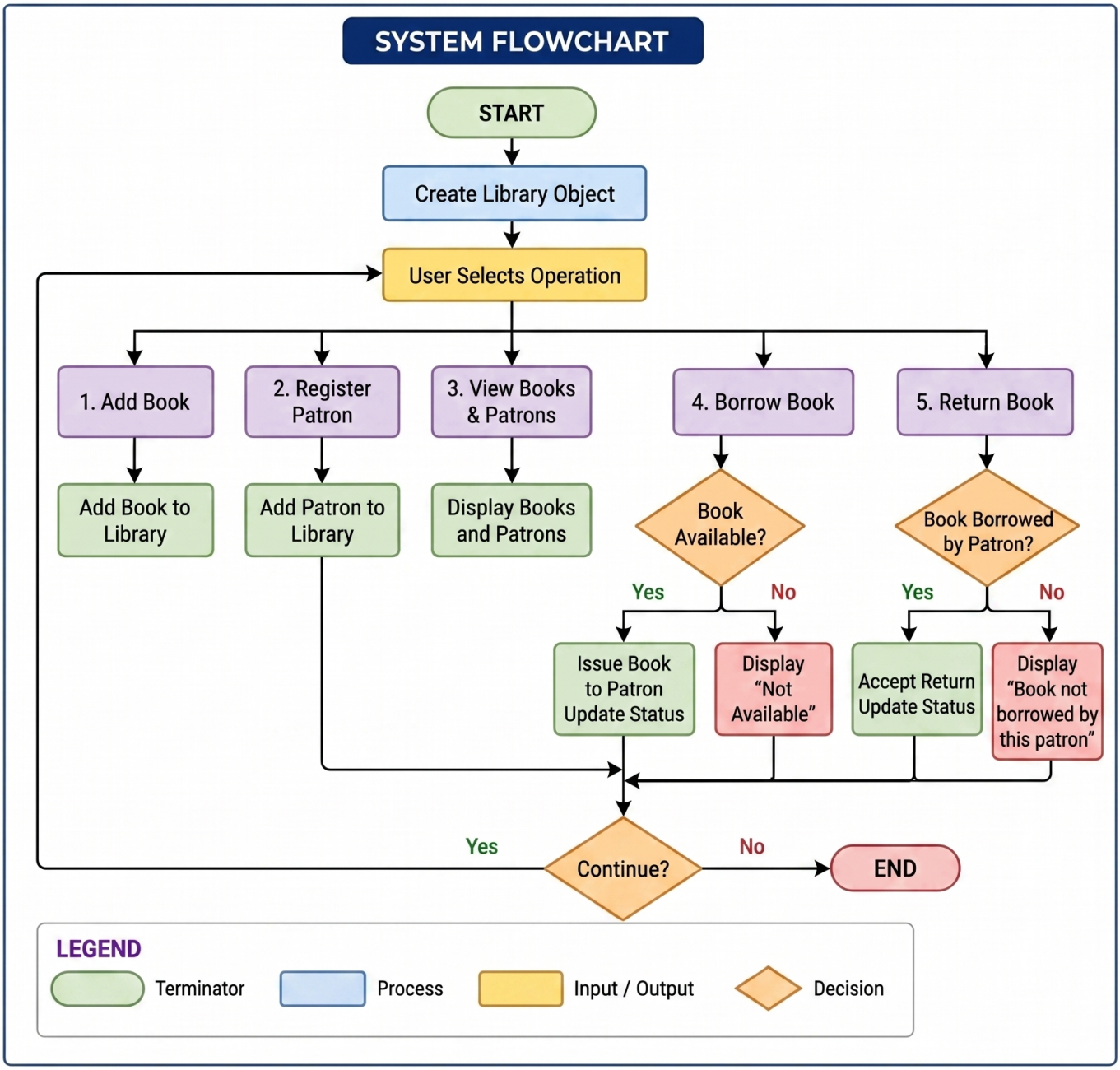
Output Screenshot 1 - Add Book, Register Patron, View Books & Patrons



# Output Screenshot 2 - Borrow Book and Return Book



## 3. System Flowchart




# 4. Algorithm

**Step 1:** Start the program.

**Step 2:** Create a Library object to maintain the records of books and patrons.

**Step 3:** Display the main menu and ask the user to choose an operation.

**Step 4:** If the user selects **Add Book**, enter the ISBN, title, and author, then store the new book in the library.

**Step 5:** If the user selects **Register Patron**, enter the patron ID and name, then save the patron's details.

**Step 6:** If the user selects **View Books & Patrons**, display all books along with their availability status and display all registered patrons.

**Step 7:** If the user selects **Borrow Book**, enter the ISBN and patron ID. Check whether the book exists and is available. If available, issue the book to the patron and update its status; otherwise, display **"Not Available."**

**Step 8:** If the user selects **Return Book**, enter the ISBN and patron ID. Verify that the book was borrowed by the same patron. If the verification is successful, accept the return and update the book status; otherwise, display **"Book not borrowed by this patron."**

**Step 9:** Ask the user whether to continue (**Yes/No**). If the answer is **Yes**, return to the main menu. If the answer is **No**, proceed to the next step.

**Step 10:** End the program.